Class 10 NCERT Maths

CLASS 10 NCERT MATHS

Chapter 11: Areas Related to Circles (Core-Basics Edition)

CHAPTER 11: AREAS RELATED TO CIRCLES

A simplified, detailed guide in Hinglish
Special edition for students who skipped Class 7-8-9

IS PDF MEIN RANG KA MATLAB:

RED text = core basic / prerequisite (Class 7-8-9 ka).
GREEN text = exam mein BAAR-BAAR aata hai (zaroor yaad rakh).
(Aur black = normal explanation.)

CORE BASICS - MISS MAT KARNA (RED PAGE)

1) CIRCLE - RADIUS, DIAMETER

Radius (r) = centre se circle tak. Diameter = 2r.

2) pi (PI) KYA HAI?

pi = circumference / diameter = 3.14 (ya 22/7).
Radius 7 ka multiple ho -> 22/7 use karo, warna 3.14.

3) AREA vs PERIMETER

Area = andar ki jagah (cm^2). Perimeter/circumference = boundary ki length (cm).

4) CIRCLE KE 2 FORMULA

Circumference = 2 pi r. Area = pi r^2.

5) ANGLE (degree)

Pura circle = 360 degree. Sector ek hissa hai jiska apna angle theta hota hai.

6) FRACTION OF CIRCLE

Sector = circle ka theta/360 hissa. Isi se area aur arc nikalte hain.

7) SQUARE / UNITS

r^2 = r x r. Area ka unit hamesha square (cm^2).

NOW: ASLI CHAPTER 11 SHURU

TOPIC 1: CIRCLE BASICS

EXAM ALERT: Ye 2 formula base hain:

Circumference = 2 pi r
Area = pi r^2

TOPIC 2: SECTOR (pizza slice)

Sector = 2 radii + arc ke beech ka area, angle theta.

EXAM ALERT: Sector formulas:

Area of sector = (theta/360) x pi r^2
Length of arc = (theta/360) x 2 pi r
Perimeter of sector = arc + 2r

TOPIC 3: SEGMENT

Segment = chord aur arc ke beech ka area.

Area of segment = Area of sector - Area of triangle

Triangle area (do side r, beech angle theta):
= (1/2) r^2 sin(theta)

TOPIC 4: MINOR vs MAJOR

Minor = chhota (angle < 180). Major = bada (angle > 180).
Major area = (total circle) - (minor area).

SOLVED EXAMPLES: HARDEST -> EASIEST

8 examples poori tarah solve karke. Upar HARDEST, neeche EASIEST.

Solved Example 1 (HARDEST) - Segment area

(Exam favourite)

Radius 10 cm, angle 90. Minor segment area? (pi=3.14)
Sector = (90/360) x 3.14 x 100 = (1/4) x 314 = 78.5
Triangle = (1/2) x 10 x 10 x sin90 = (1/2)(100)(1) = 50
Segment = 78.5 - 50 = 28.5 cm^2.

Solved Example 2 - Sector area (22/7)

(Exam favourite)

Radius 21 cm, angle 60. Sector area? (pi=22/7)
= (60/360) x (22/7) x 21 x 21
= (1/6) x 22 x 3 x 21
= (1/6) x 1386 = 231 cm^2.

Solved Example 3 - Arc length

Radius 7 cm, angle 90. Arc length? (pi=22/7)
= (90/360) x 2 x (22/7) x 7
= (1/4) x 44 = 11 cm.

Solved Example 4 - Area of circle

Radius 14 cm. Area? (pi=22/7)
= (22/7) x 14 x 14 = 22 x 28 = 616 cm^2.

Solved Example 5 - Circumference

Radius 7 cm. Circumference? (pi=22/7)
= 2 x (22/7) x 7 = 44 cm.

Solved Example 6 - Sector perimeter

Radius 7, angle 90. Sector perimeter? (pi=22/7)
  Arc = (1/4) x 44 = 11. Perimeter = arc + 2r
      = 11 + 14 = 25 cm.

Solved Example 7 - Find radius from area

Area = 154 cm^2. Radius? (pi=22/7)
  pi r^2 = 154 -> (22/7) r^2 = 154
  r^2 = 154 x 7 / 22 = 49 -> r = 7 cm.

Solved Example 8 (EASIEST) - Diameter

Radius 9 cm. Diameter? -> 2r = 18 cm.

Q AND A TIME

Q1. CORE: circumference aur area ke formula likho.

Q2. Radius 14 cm circle ka area aur circumference.
    (pi=22/7)

Q3. Radius 6 cm, angle 60. Sector area. (pi=3.14)

Q4. Radius 14 cm, angle 90. Arc length. (pi=22/7)

Q5. Radius 4 cm, angle 90. Minor segment area.
    (pi=3.14, sin90=1)

Q6. Sector perimeter ka formula kya hai?

Q7. CORE CHECK: kab 22/7 use karte hain, kab 3.14?

SUMMARY

---

1. Circumference = 2 pi r ; Area = pi r^2.

2. Sector area = (theta/360) pi r^2.

3. Arc length = (theta/360) 2 pi r.

4. Segment = sector - triangle
   = (theta/360) pi r^2 - (1/2) r^2 sin(theta).

5. pi = 22/7 (radius 7 ka multiple) ya 3.14.

CORE (RED page) revise: radius/diameter, pi, area vs perimeter, fraction of circle (theta/360), units.
